Haocheng Cai

Department of Mathematics, University of Washington
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Education

University of Washington **2020–2027 (expected)**
Ph.D. in Mathematics
Advisor: Sándor Kovács

University of California, Los Angeles **2016–2020**
B.S. in Mathematics, Specialization in Computing, *cum laude*

Research Interests

Commutative Algebra, Algebraic Geometry, and Noncommutative Algebra

Publications and Preprints

1. **Haocheng Cai**. "S_r Properties of Generalized Cohen–Macaulay Modules and Rings Having Liftable Local Cohomology via Serre Depth." arXiv:2607.08995.
2. **Haocheng Cai**. "Big Buchsbaum Module and Perfection." *In preparation.*

Seminar Series

Noncommutative Rings 2021
University of Washington
Delivered a year-long biweekly presentation series based on I. N. Herstein's *Noncommutative Rings*.

D-Module Theory 2024
Noncommutative Algebra Seminar, University of Washington
Led a two-quarter biweekly seminar series on foundational topics in D-module theory and dimension theory.

Selected Talks

Introduction to D-Modules May 2023
1-2-3 Seminar, University of Washington

The Riemann–Roch Theorem and Its Applications Winter 2025
Algebraic Curves Reading Group, University of Washington

Iitaka Dimension Spring 2025
General Exam Talk, University of Washington

Measuring Singularities in Characteristic 0 and p via Thresholds February 2026
1-2-3 Seminar, University of Washington

Uses of Non-Finitely Generated Modules Fall 2026 (scheduled)
UW Back to School Seminar, University of Washington

Serre Depth and Its Applications Fall 2026 (scheduled)
Student Algebraic Geometry Seminar, University of Washington

AI for Mathematics and Formalization

Nagata's Criterion for UFDs Winter 2025
Independent Formalization Project

Monogenic Extensions of Regular Local Rings Spring 2025
Mentor, UW Math AI Lab

Teaching Experience

Instructor of Record *University of Washington*

MATH 124 A: Calculus with Analytic Geometry I	Summer 2022
MATH 381: Discrete Mathematical Modeling	Summer 2025

Teaching Assistant *University of Washington*

MATH 124: Calculus with Analytic Geometry I	Fall 2021
MATH 125: Calculus with Analytic Geometry II	Winter 2022
MATH 327: Introduction to Real Analysis	Spring 2022
MATH 124: Calculus with Analytic Geometry I	Fall 2022
MATH 403: Introduction to Modern Algebra II	Winter 2023
MATH 403: Introduction to Modern Algebra II	Spring 2023
MATH 300: Introduction to Mathematical Reasoning	Fall 2023
MATH 381: Discrete Mathematical Modeling	Winter 2024
MATH 327: Introduction to Real Analysis	Spring 2024
MATH 411: Introduction to Modern Algebra with Applications I	Fall 2024
MATH 402: Introduction to Modern Algebra I	Winter 2025
MATH 412: Introduction to Modern Algebra with Applications II	Winter 2025
MATH 300: Introduction to Mathematical Reasoning	Spring 2025
MATH 208: Linear Algebra	Fall 2025
MATH 208: Linear Algebra	Spring 2026
MATH 125: Calculus with Analytic Geometry II	Fall 2026